

1 AIP2 - GUI Configuration

1.1 Overview

The layout for the new GUI of the AIP2 terminal (tile design) is stored in XML files. XML files can be edited using a standard text editor. Microsoft's *XML Notepad 2007* can also be used. This XML editor provides a clearer presentation, a user-friendly copy function for entire objects and the possibility to move complete objects. *XML Notepad 2007* was used for the generation of screenshots included in this document.



In the configuration, font sizes and positions are given in points in relation to a screen resolution of 600 points in height. Values must be scaled and then rounded to whole dots when using a screen with a higher resolution. Therefore, the proportions can slightly vary depending on the screen resolution.



Colors are specified in XML files in a reversed RGB notation (**Blue/Green/Red** instead of **Red/Green/Blue**). If the entry is in the hexadecimal format, the two places behind the symbol \$ define the color blue, the next two places the color green and red is defined by the last two places.

1.2 Filing XML files in the server

Like the INI files *ctaiplay.ini* and *ctaipbut.ini*, the XML files that define the GUI are located on the server in the sub directory *ctnet\win\aip2*. XML files are filed in the sub directory *gui*.

1.2.1 Scope Concept

The INI files in the subdirectory *<SystemNo>\custom\aip2* and the XML files in the subdirectory *<SystemNo>\custom\aip2\gui* can be overridden customer-specifically in deviation from the standard. Various *scopes* are provided in order that the different changes do not overwrite each other.

Scope	Directory	Examples
Standard	<i>ctnet\win\aip2\<x>.<y></i>	<i>ctnet\win\aip2\ctaiplay.ini</i> <i>ctnet\win\aip2\gui\l_anr.xml</i>
Standard scope	<i><SystemNo>\custom\aip2\<x>.<y></i>	<i>1\custom\aip2\ctaiplay.ini</i> <i>1\custom\aip2\gui\l_anr.xml</i>
Custom Scope	<i><SystemNo>\custom\aip2\<x>@custom.<y></i>	<i>1\custom\aip2\ctaiplay@custom.ini</i> <i>1\custom\aip2\gui\l_anr@custom.xml</i>
VAR scope	<i><SystemNo>\custom\aip2\<x>@var.<y></i>	<i>1\custom\aip2\ctaiplay@var.ini</i> <i>1\custom\aip2\gui\l_anr@var.xml</i>
Local scope	<i><SystemNo>\custom\aip2\<x>@local.<y></i>	<i>1\custom\aip2\ctaiplay@local.ini</i> <i>1\custom\aip2\gui\l_anr@local.xml</i>

Standard scope

The Standard Scope is used to create a copy of the standard. This ensures that changes to the standard, sometimes supplied with a service pack, do not interfere with the collection terminal and therefore, the collection software remains unchanged. After generating the copy, the required changes to the standard must be either copied again or synchronized. The changes can then be integrated into the Standard Scope.

Custom Scope

The Custom Scope is reserved for MPDV to file customer specific configurations. A file is stored in the Custom Scope if the file name includes **@custom** before the extension.

VAR scope

The VAR Scope is reserved for partners (Value Added Reseller) to store changes for customers of partners. A file is stored in the VAR scope if **@var** is inserted before the extension.

Local scope

The Local Scope is reserved for customers to store their own customized files. A file is stored in the Local Scope if **@local** is inserted before the extension.

The priority of the different scopes is ascending from the standard scope to the local scope. A file in the local scope takes priority over a file included in the standard scope.

INI and CFG files are processed differently to XMLfiles:

INI and CFG files are merged per section, that means sections in the standard are totally replaced by potentially existent sections deriving from individual scopes. Settings in the Local Scope have highest priority as they are processed at last and therefore overwrite settings from the scope located above.

XML files are not merged but accepted. That means only files are processed located in the list of scopes at the bottom.



The only exception is the file *globaldefines.xml*. The content of that file is merged with the settings of the individual scopes. It is therefore possible to overwrite individual settings (i.e. font size or color) without copying the complete file. If you would like to overwrite a certain element of the file *globaldefines.xml*, please copy the file, delete all elements to be accepted from the standard file and then store the file in the relevant Scope.

1.2.2 Specific layouts of terminals or terminal groups

Like the INI files *ctaipplay.ini* and *ctaipbut.ini*, the XML files that define the GUI are stored on the HYDRA server in the subdirectory `<SystemNo>\custom\aip2`. Standard XML files are filed in the sub directory *gui*.

If different GUI layouts are required for specific terminals or terminal groups, the non-standard XML files are stored in the following subdirectories:

Reference	Subdirectory	Example
Terminal group	tgrp_<Terminal group>\gui	tgrp_900\gui
Terminal	tnr_<Terminal number>\gui	tnr_100\gui

XML files stored in these sub directories replace standard files with the exception of the *globaldefines.xml* file whose content is merged with the relevant standard files.

1.2.3 Loading configuration files during restart

Every time the AIP2 is started, INI, CFG and XML files are updated from the server and automatically activated in the terminal.

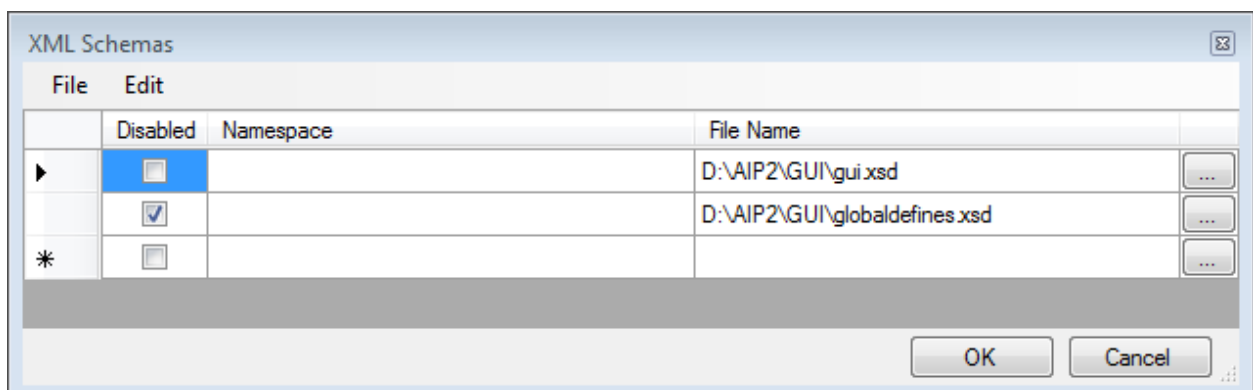


When changing the layout, please note that the changed layouts are not overwritten when the AIP is started. Updating of configuration files can be deactivated in the file *ctaip.ini* by adding the parameter *SkipAipStartupUpdate* to the entry *parameters=* in the section *[system]*.

1.2.4 Syntax check via XML Schema Definition (XSD)

An XML Schema Definition (XSD) defines the structure of an XML file. Depending on the editor used, a syntax check of the edited XML file is performed. In addition, you can select the value of specific fields via a selection list.

In the *XML Notepad 2007* of *Microsoft*, you can enter XML Schema Definitions via the menu item *View – Schemas...* and enable or disable them:

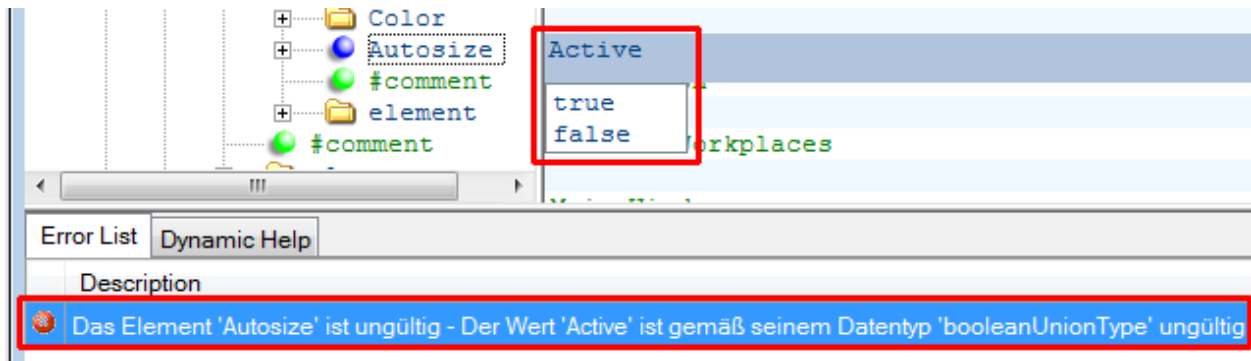


The file *globaldefines.xsd* includes the schema for the file *globaldefines.xml*. The file *gui.xsd* includes the schema for the other XML files used for the GUI configuration. The two XSD files are located in the HYDRA server in the same directory as the corresponding XML files.



The two XSD files *globaldefines.xsd* and *gui.xsd* are not compatible. For this reason, one of these files must always be *disabled*.

The following example shows a syntax check and a selection list:



1.3 Settings

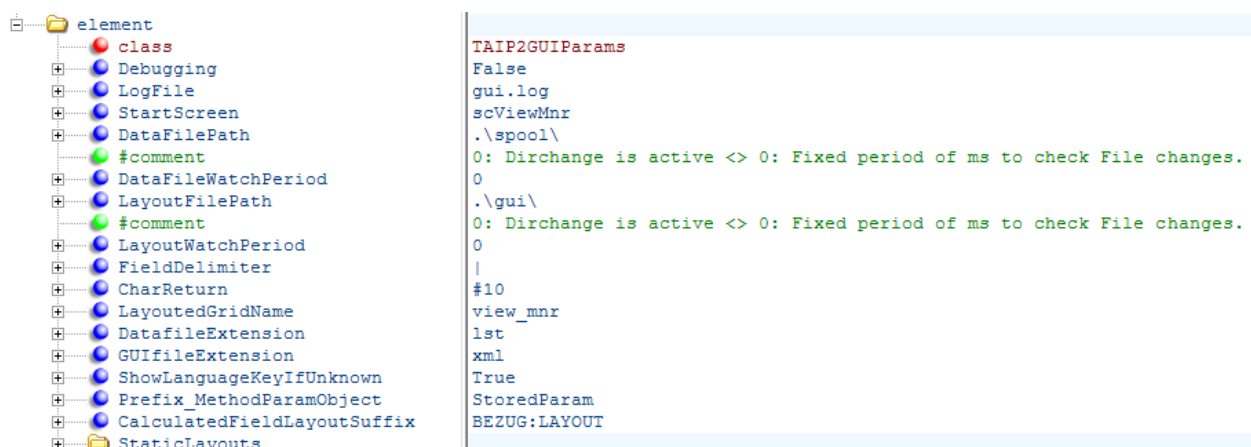
The file *globaldefines.xml* includes general settings, constants, data sources, calculated fields and functions.



Changes in the file *globaldefines.xml* are only active after restart of the AIP2.

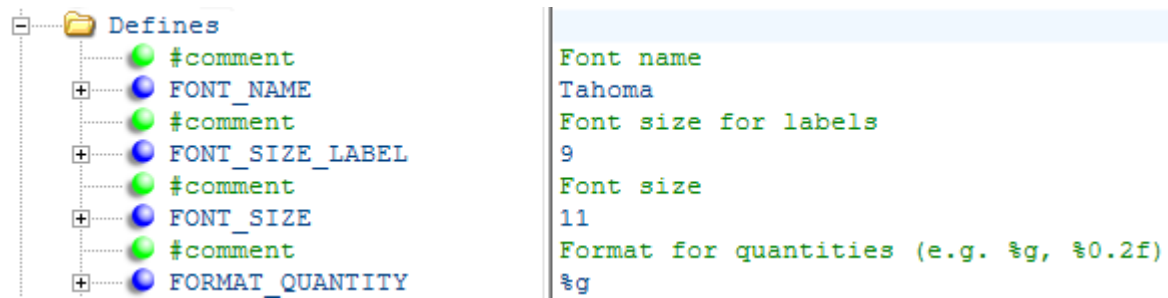
1.3.1 General settings

Standard settings control program processing and may not be changed.



1.3.2 Constants (Defines)

The section *Defines* specifies the Constants used for different layout configurations. For example, you can change color or font size at a central location using these constants.



The following constants can be set:

FONT_NAME

This constant sets the font of the GUI. The standard setting is „Tahoma“.

FONT_SIZE_LABEL

This constant sets the font size for the GUI labeling. The standard setting is 8.

FONT_SIZE

The constant FONT_SIZE sets the font size for the displayed data in the GUI. The standard setting is 10.

FONT_SIZE_HEADING

FONT_SIZE_HEADING defines the font size for the headings on the right hand side. The standard setting is 10.

FORMAT_QUANTITY

This constant sets the format for the display of quantities.

Standard setting is „%g“.

„%g“: Automatically formats the shortest display: If the quantity is an integer, then without decimal places and without decimal separators. If the quantity is not an integer, the existing decimal places are output after a decimal separator (maximum 15 decimal places). No thousand separator is output.

„%0.0f“: No decimal places, without thousands separator.

„%0.2f“: Always two decimal places, without thousands separator.

„%0.2n“: The format n is the same as the format f, but the resulting string contains thousands separators, if a thousands separator is configured in the regional settings.



The set format only affects configured layouts and not dynamic dialogs. A delimiter for thousands is not available for dynamic dialogs.

FORMAT_CYCLE

FORMAT_CYCLE defines the output format for cycle times (target cycle and actual cycle) if they are not output as durations (as in the standard system) but in seconds. Formatting is issued like in the previous field. Standard setting is "%0.3f". Please refer to section "1.2.4 CalculatedFields" for further information.



The set format only affects configured layouts and not dynamic dialogs.

COLOR_MENU_BUTTON

This constant defines the background color for specific buttons on the left hand side, like the button "<back" and "PZE". For example the button "< Back" and the button "PZE" on the start page belong here. Standard setting is "\$C0C0C0" (light gray).

COLOR_MENU

This constant controls the background color of the buttons used for selecting workplaces in the "Home"-page and for calling functions if an object was selected. Standard setting is "\$E0E0E0" (lighter gray).

COLOR_MENU_ACTIVE

This constant can set a background color for the selected workplace. Standard setting is "\$909090" (gray).

COLOR_BACKGROUND

This constant can set a background color to display data on the right hand side. The color should correspond with the COLOR_MENU_ACTIVE. Standard setting is "\$909090" (gray).

COLOR_MARGINS

This constant sets the color for the borders of the layout. Standard setting is "\$FFFFFF" (white).

COLOR_HEADING

This constant defines the background color for the upper headings on the right hand side. Standard setting "\$833014" (dark blue).

COLOR_HEADING_2

This constant defines the background color for the headings in the middle on the right hand side. Standard setting is "\$974428" (blue).

COLOR_HEADING_3

This constant defines the background color for the lower headings on the right hand side. Standard setting is "\$AA583B" (light blue).

COLOR_FONT_HEADING

This constant defines the font color of the headings on the right hand side. Standard setting is "\$FFFFFF" (white).

COLOR_TILE

This constant defines the background color of the individual tiles on the right hand side. Standard setting is "\$F8F8F8" (very light gray).

COLOR_FONT

COLOR_FONT sets the standard font color. Standard setting is "\$202020" (dark gray).

COLOR_STATUS_PRODUCTION

This constant defines the color for the status *production*. Standard setting is "1077248" (dark green, \$107000). With this constant, the color must be entered as a decimal value to avoid the display in a different color on specific screens.

COLOR_STATUS_NO_PRODUCTION

This constant controls the color for the all statuses except *production*. Standard setting is "\$1090FF" (dark yellow).

COLOR_STATUS_NOT_ASSIGNED

This constant defines the color for the status "Not assigned". Standard setting is "\$1010D0" (dark red).

COLOR_YIELD

This constant defines the color to display yields. Standard setting is "1077248" (dark green, \$107000). With this constant, the color must be entered as a decimal value to avoid the display in a different color on specific screens.

COLOR_SCRAP

COLOR_SCRAP controls the color to display scrap. Standard setting is "\$1010D0" (dark red).

COLOR_INSPECTION_DUE

This constant can set a background color to display a CAQ inspection due. Standard setting is "\$1090FF" (yellow).

COLOR_INSPECTION_DONE

This background color indicates if a minimum inspection scope was reached. Standard setting is "1077248" (dark green, \$107000). With this constant, the color must be entered as a decimal value to avoid the display in a different color on specific screens.

COLOR_INSPECTION_ERROR

If an error occurs in the inspection planning, the relevant area is shown in the color set. Standard setting is "\$1010D0" (dark red).

COLOR_MAINTENANCE_STATUS_0, ..., COLOR_MAINTENANCE_STATUS_3

Color to display maintenance status of resources. Standard settings are "\$FFFFFF" (white), "\$873418" (blue), "\$1090FF" (yellow) and "\$1010D0" (red).



The name of customer specific constants must include the prefix "U". This way, they cannot be mixed up with constants of the standard.

The following example shows how to override a constant in a scope so that it is merged with the settings in the standard scope.



1.3.3 Data sources (ProviderDefinition)

The settings define the correlation between the individual data sources and may not be changed.

1.3.4 Calculated fields

Calculated fields configure the display of the target and actual cycle. Both fields can be provided either as "time for 1000 pieces", "time for one piece" or "pieces per minute". The setting is done using the calculated fields SZY_CALC and IZY_CALC. The following formulas can be stored in the attribute *Expression* :

Field	Presentation	Formula
Target cycle	Time for 1000 pieces	FloatToVar(FieldValue('SZY', AsDouble, 0))
	Time for one piece	FloatToVar(FieldValue('SZY', AsDouble, 0) / 1000)
	Pieces per minute	FloatToVar(60000 / FieldValue('SZY', AsDouble, 0))
Actual cycle	Time for 1000 pieces	FloatToVar(FieldValue('IZY', AsDouble, 0))
	Time for one piece	FloatToVar(FieldValue('IZY', AsDouble, 0) / 1000)
	Pieces per minute	FloatToVar(60000 / FieldValue('IZY', AsDouble, 0))

Three options can be used as comments in the file globaldefines.xml:

element	
class	TScriptDefinition
ID	SZY_CALC
Identifier	SZY_CALC
Provider	MNR
#comment	Seconds per 1000 pieces: FloatToVar(FieldValue('SZY', AsDouble, 0))
#comment	Seconds per piece: FloatToVar(FieldValue('SZY', AsDouble, 0) / 1000)
#comment	Pieces per minute: FloatToVar(60000 / FieldValue('SZY', AsDouble, 0))
Expression	FloatToVar(FieldValue('SZY', AsDouble, 0) / 1000)
#comment	IZY_CALC - Actual cycle
element	
class	TScriptDefinition
ID	IZY_CALC
Identifier	IZY_CALC
Provider	MNR
#comment	Seconds per 1000 pieces: FloatToVar(FieldValue('IZY', AsDouble, 0))
#comment	Seconds per piece: FloatToVar(FieldValue('IZY', AsDouble, 0) / 1000)
#comment	Pieces per minute: FloatToVar(60000 / FieldValue('IZY', AsDouble, 0))
Expression	FloatToVar(FieldValue('IZY', AsDouble, 0) / 1000)

The formatting of both fields is configured using the constant `FORMAT_CYCLE`. Please refer to section "1.2.2 Constants (Defines)" for further information.

1.3.5 Functions (ScriptDefinitions)

The settings may not be changed during configuration of the GUI.

Fields that start a function are identified by the attributes *Extention* and *ScriptName*. The following example controls the height of the entries in the list of workplaces in the main view (a_list_mnr.xml):

control	
Left	0
Height	
Extention	MakeDataSensitive
ScriptName	HeightListMNR
#text	52
Width	210

The entry in the field `#text` is not active in this case. It is overwritten by the result of the previously entered function. If you want to change the height entered in this field, you must delete the two attributes *Extention* and *ScriptName*. Please note that in this case the data dependent identification of the height is disabled.

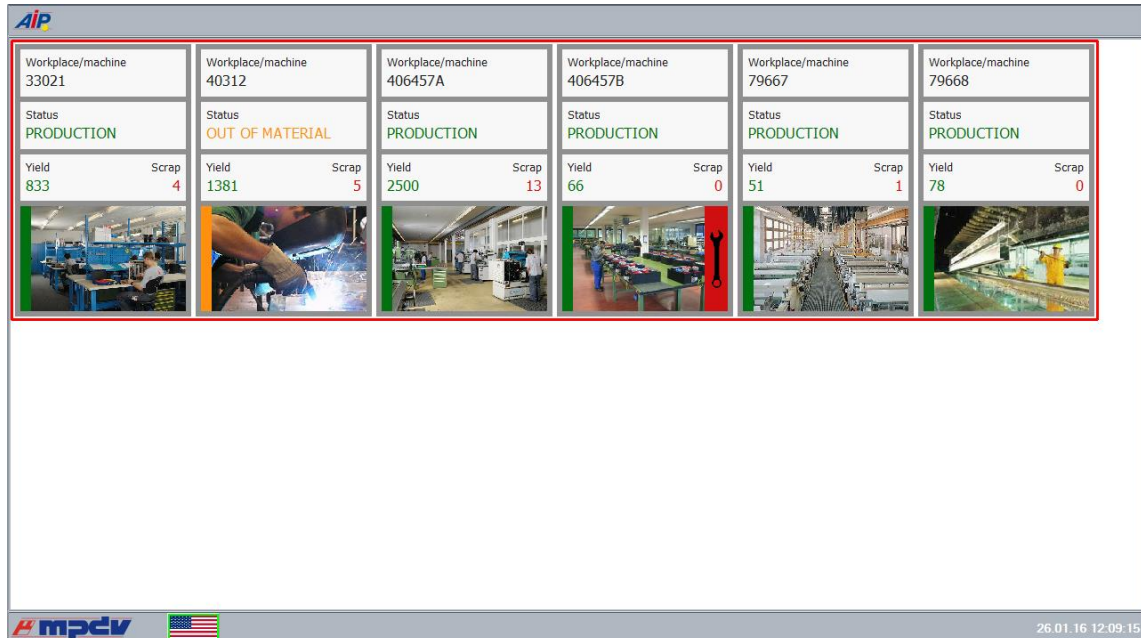
1.4 Layout definition

The definition of the layout is separated into layout files beginning with „l_“ and areas consisting of file names beginning with "a_".

There are the following layouts and areas in the standard:

I_view_mnr.xml

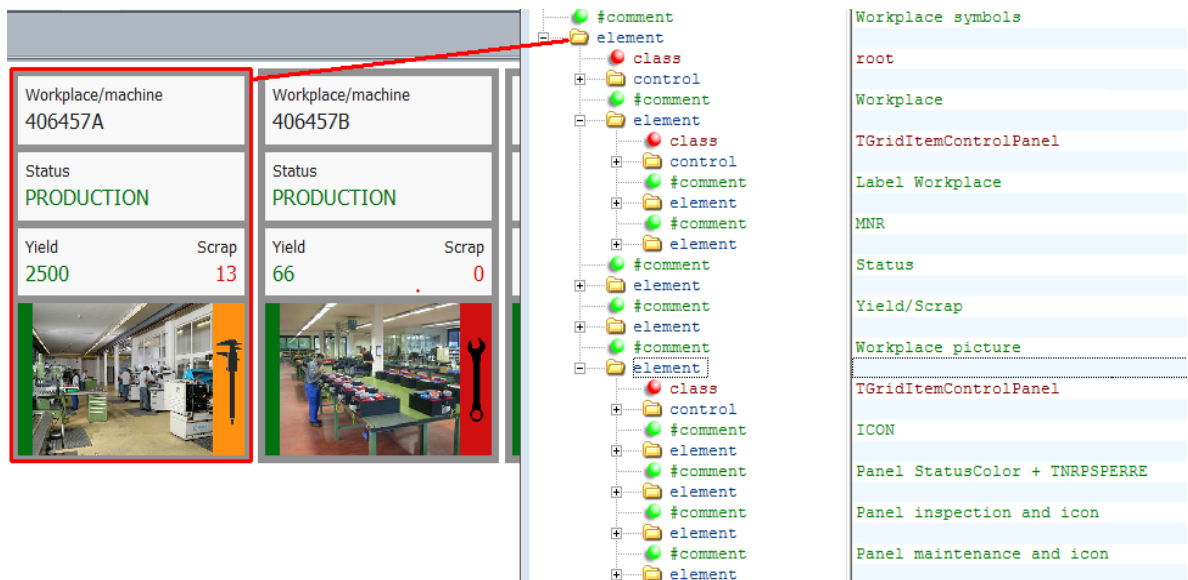
The tile view shows workplaces assigned to the terminal:



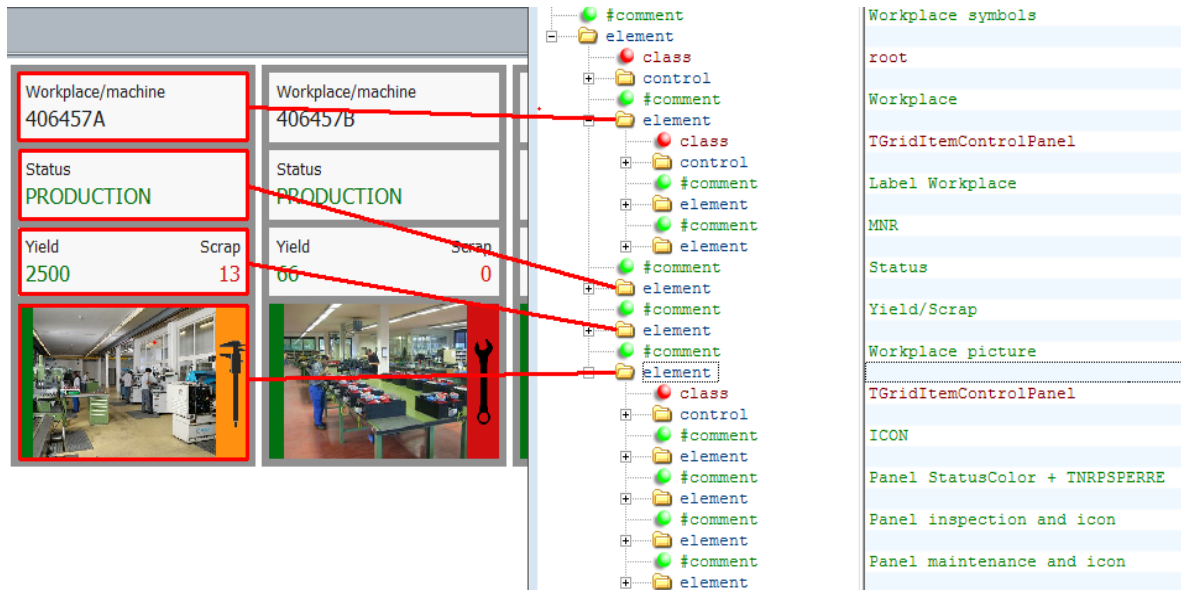
The structure for the individual workplaces is stored in the file **a_view_mnr.xml**.

The following screenshots show the assignment of the elements in the file to the objects in the GUI.

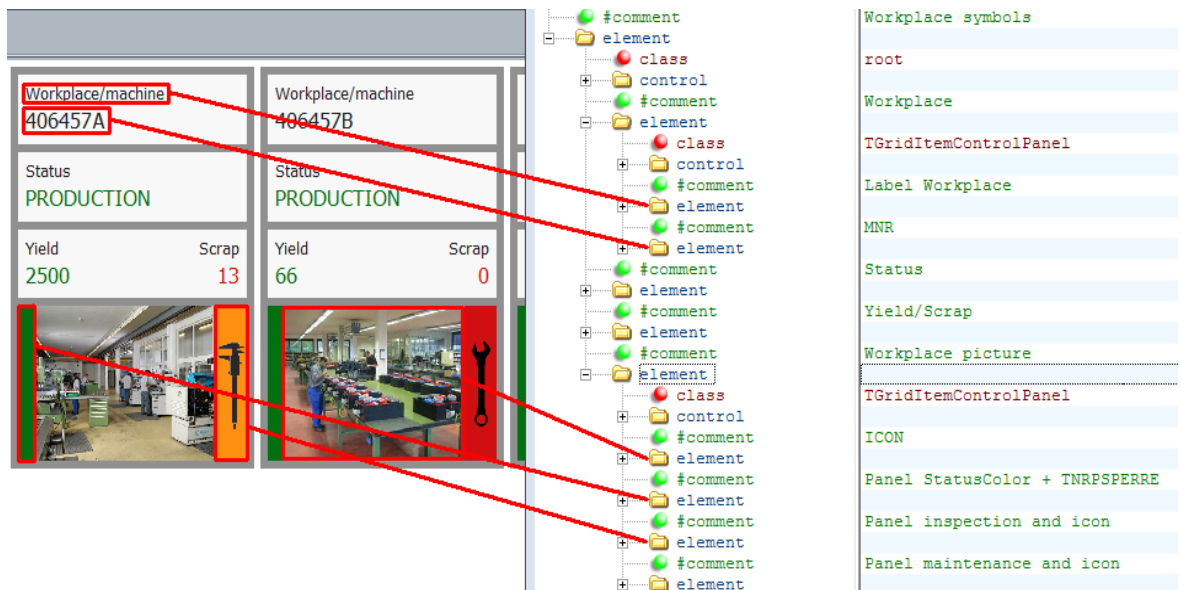
The element on the highest level of the tree structure is the big outer tile with gray frame:



The elements in the next level of the tree structure are the 3 tiles in light gray and the tile including the image:



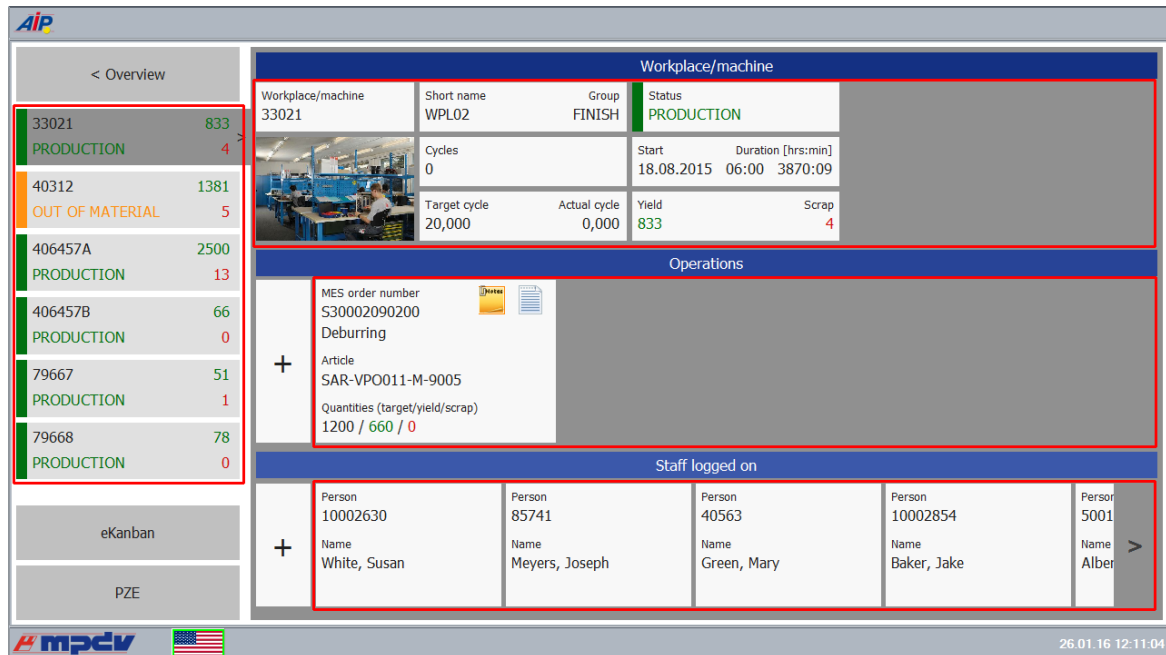
The elements in the next level of the tree structure are assigned as shown in the example of the light gray tile on top and the tile including the image at the bottom:



The lowest element in the list defines the layout of the tile including the screwdriver.

I_main.xml

Select a workplace to reach the main view:



The file **a_list_mnr.xml** contains the structure of the view of a workplace located in the list on the left hand side.

The file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace. This area is used for all layouts that follow.

The file **a_list_anr.xml** stores the layout of an operation in the middle of the screen on the right hand side. The button to the left, which is used to log on an operation, as well as all other buttons outside a red frame are located directly in the **I_main.xml** layout.

Various lists can be displayed at the bottom on the right hand side. You can set in the workplace configuration which of the 3 lists are available at a workplace. The displayed data are defined in the following files:

- **a_list_pnr.xml**: Persons logged on
- **a_list_pnrg.xml**: Persons logged on at a group workplace
- **a_list_res.xml**: Resources logged on
- **a_list_emat.xml**: Logged on input material
- **a_list_amat.xml**: Produced output batches

I_mnr.xml

If you click on the area showing the data for the selected workplace, you will get to the workplace layout:

Workplace/machine					
Workplace/machine 33021	Short name WPL02	Group FINISH	Status PRODUCTION		
	Cycles 0	Start 18.08.2015 06:00		Duration [hrs:min] 3870:12	
	Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4	

As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The buttons on the left side are located in the layout **I_mnr.xml**.

I_anr.xml

If you click an operation in the main view, the operation layout appears:

Workplace/machine					
Workplace/machine 33021	Short name WPL02	Group FINISH	Status PRODUCTION		
	Cycles 0	Start 18.08.2015 06:00		Duration [hrs:min] 3872:15	
	Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4	

Operation			
MES order number S30002090200 Deburring	Quantities (target/yield/scrap) 1200 / 660 / 0 55%	Comments	
Article SAR-VPO011-M-9005 Mirror housing right, black	Since logon (target/yield/deviation) / 0 / %	Planned duration 6:40	Partitioning 1

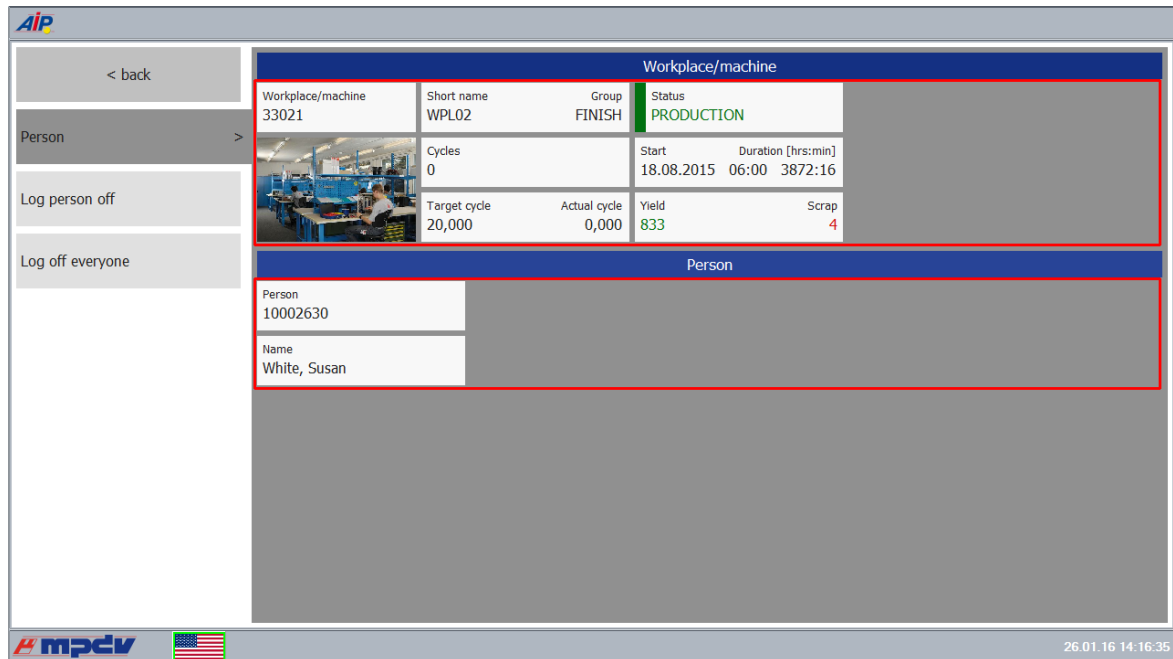
As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The file **a_data_anr.xml** contains the definition for the area in the middle of the layout showing data for the operation.

The buttons on the left side are located in the layout **I_anr.xml**.

I_pnr.xml

The layout for staff appears if you select a person in the main view



Workplace/machine				
Workplace/machine 33021	Short name WPL02	Group FINISH	Status PRODUCTION	
	Cycles 0	Start 18.08.2015 06:00	Duration [hrs:min] 3872:16	
	Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4

Person	
Person 10002630	
Name White, Susan	

As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

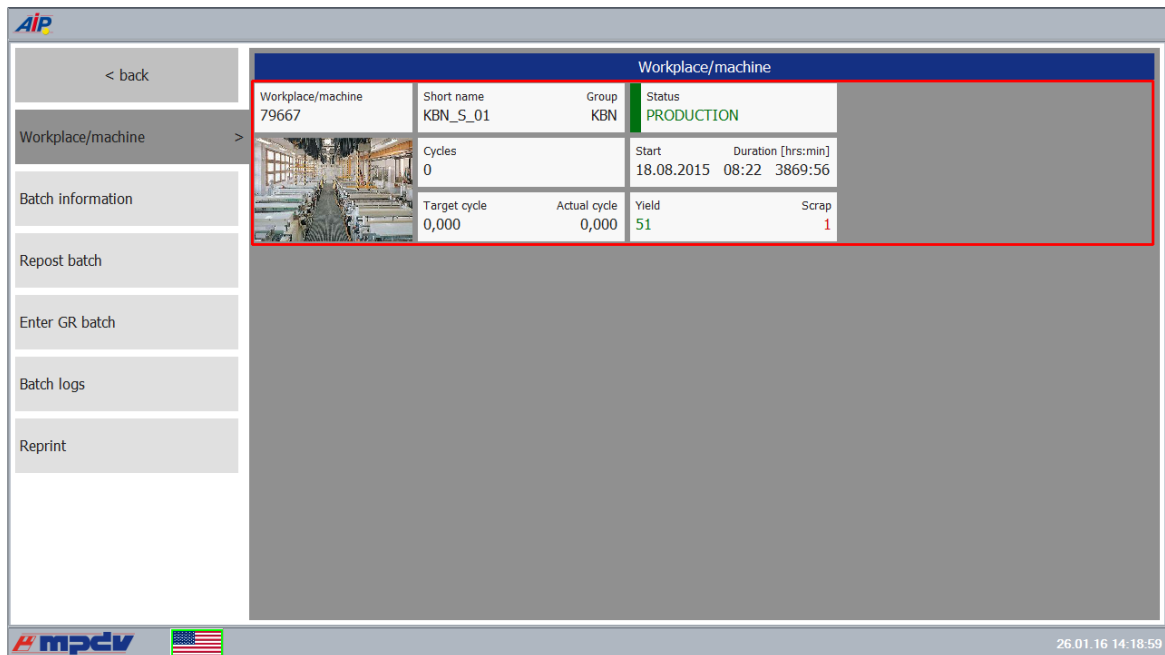
The file **a_data_pnr.xml** contains the definition for data relating to staff in the middle of the layout.

The button on the left side is located in the layout **I_pnr.xml**.

This layout is also used to display data and to request functions for staff logged on to a group workplace. When requesting the functions, an error message appears as staff and operations are logged on together to a group workplace.

I_res.xml

The resource layout opens if in the main view the third list "Resources logged on" is displayed and you click a resource:



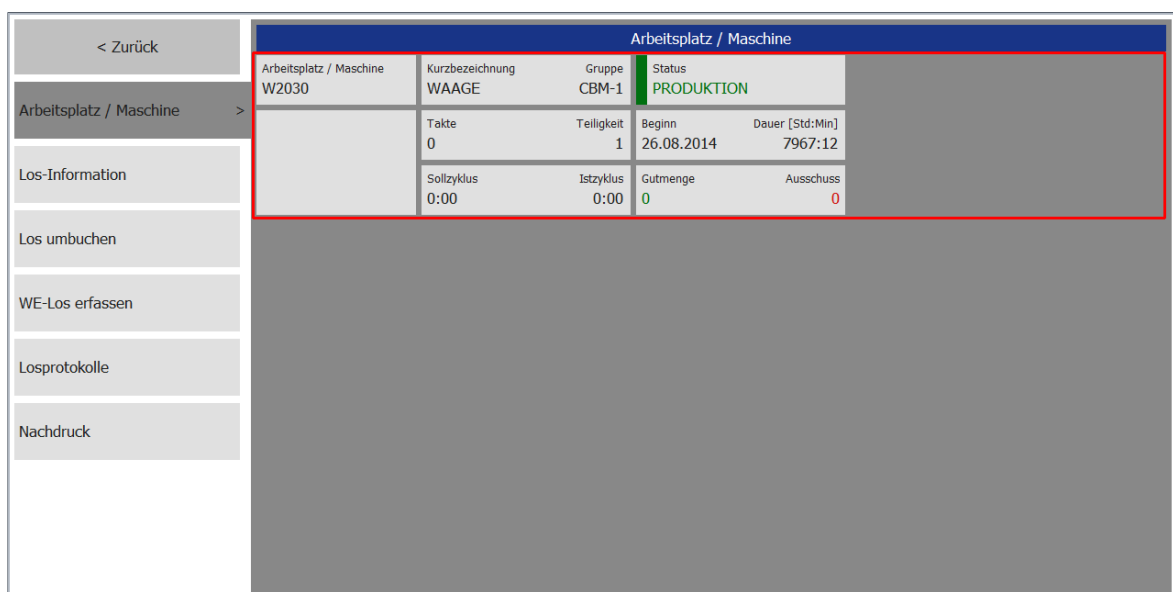
As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The file **a_data_res.xml** contains the definition for data relating to a resource in the middle of the layout.

The buttons on the left side are located in the layout **I_res.xml**.

I_mat.xml

To request material layout, go to the main view. Click the button containing three dots to the left of the 3. lists "Input material logged on" and "Produced output batches":



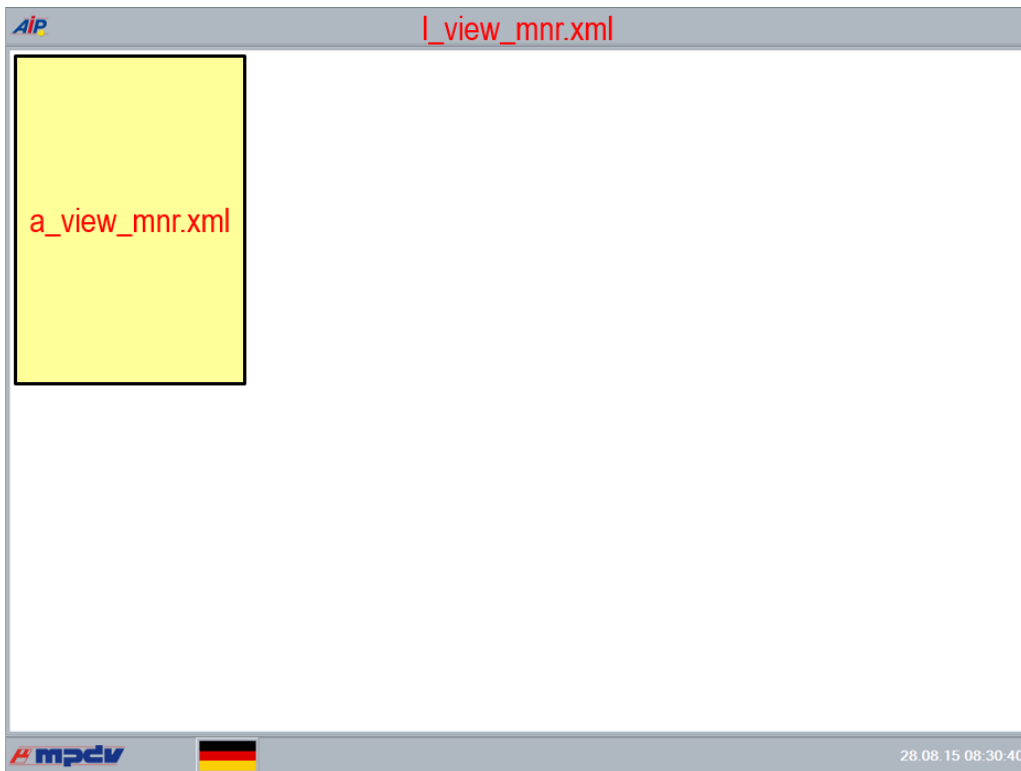
As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The buttons on the left side are located in the layout **I_mat.xml**.

1.4.1 Overview XML files

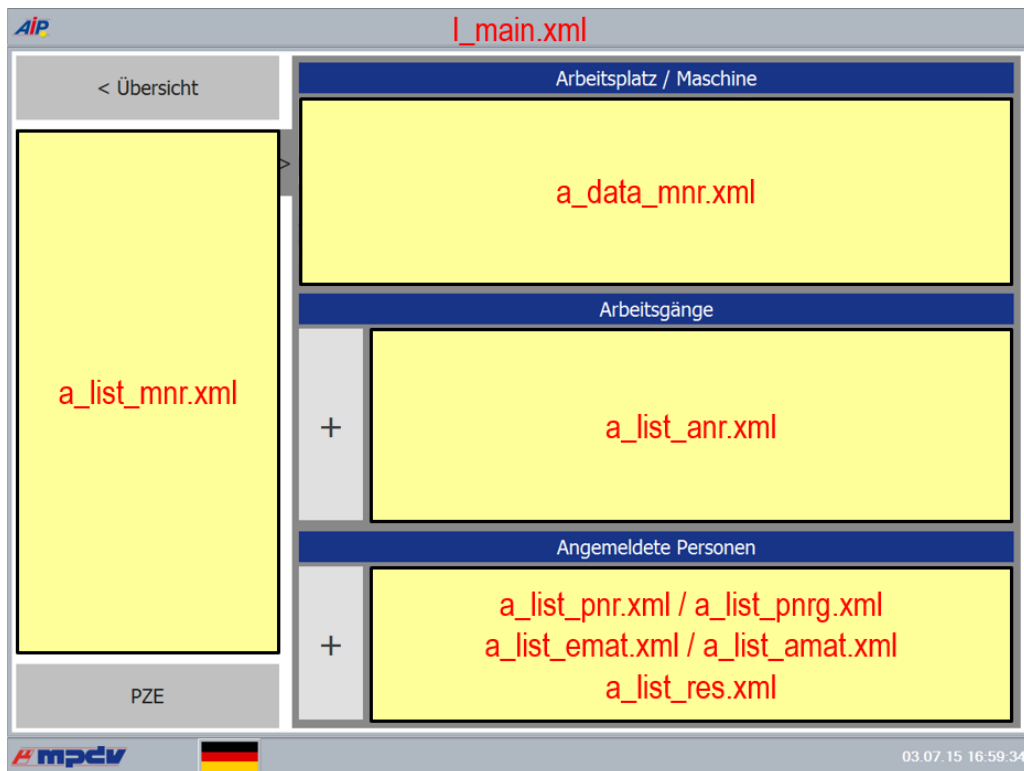
Icon

view:



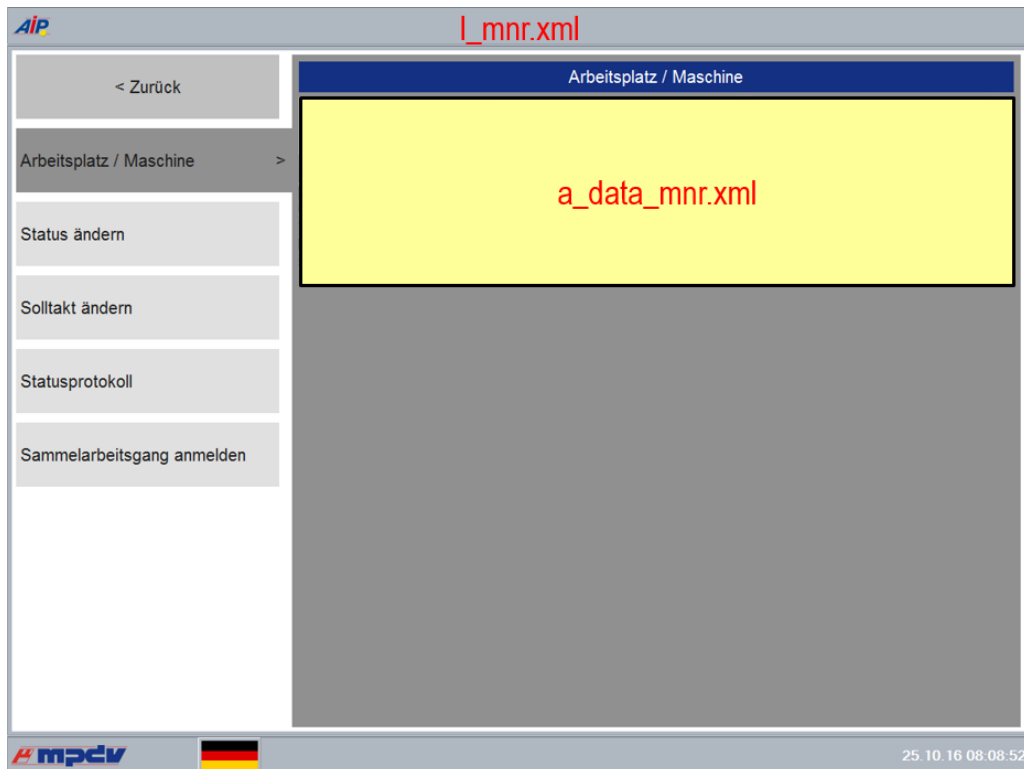
Main

view:



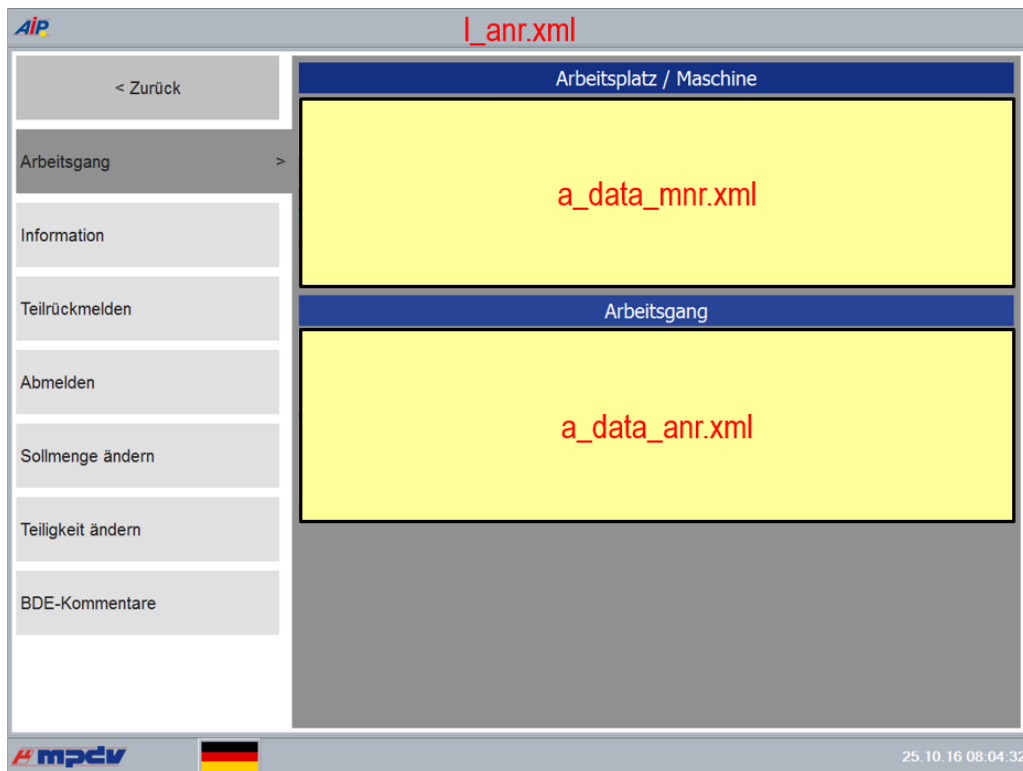
Workplace

layout:



Operation

layout:



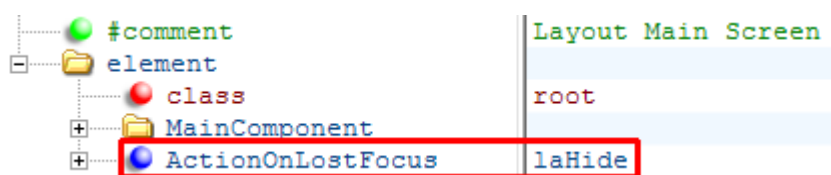
1.4.2 Layouts depending on the worplace type

You can override the layouts requested in the main view depending on the *batch management* and the *workplace type*. Both settings are stored in the dialog *Workplace and resource configuration* in the tab *Workplace configuration* and consist of one letter. If you copy a layout and both letters are written as lower case letters, are separated by an underscore (" _") and added to the file name, then this layout is used for all workplaces including batch management and used with the relevant workplace type.

For example, buttons should be made available with other dynamic dialogs for order postings at a packing station (letter "C") without batch management (letter "N"). To do so, copy the layout **I_anr.xml** onto the file name **I_anr_nc.xml**. You can then modify the buttons for the packing station in this layout.

1.4.3 Taking over changes in the layout configuration

Using the attribute „ActionOnLostFocus“, you can make the following settings per layout:



laFree

Using this setting, the displayed layout is discarded on changing to another layout and loaded anew from the XML files on the next start of this layout. Changes in the configuration of this layout, which were made in the meantime, are taken over.

laHide

With this setting, the layout is not discarded, but stays in the background when you change to another layout. Changes of the layout configuration, that were saved after the first layout display, do not have an effect as the layout is not loaded anew from the XML files.

The setting *laHide* is applied by default in the layout of the main view (**I_main.xml**) to keep the scroll position in the lists on the right hand side when you return to this layout from another layout.

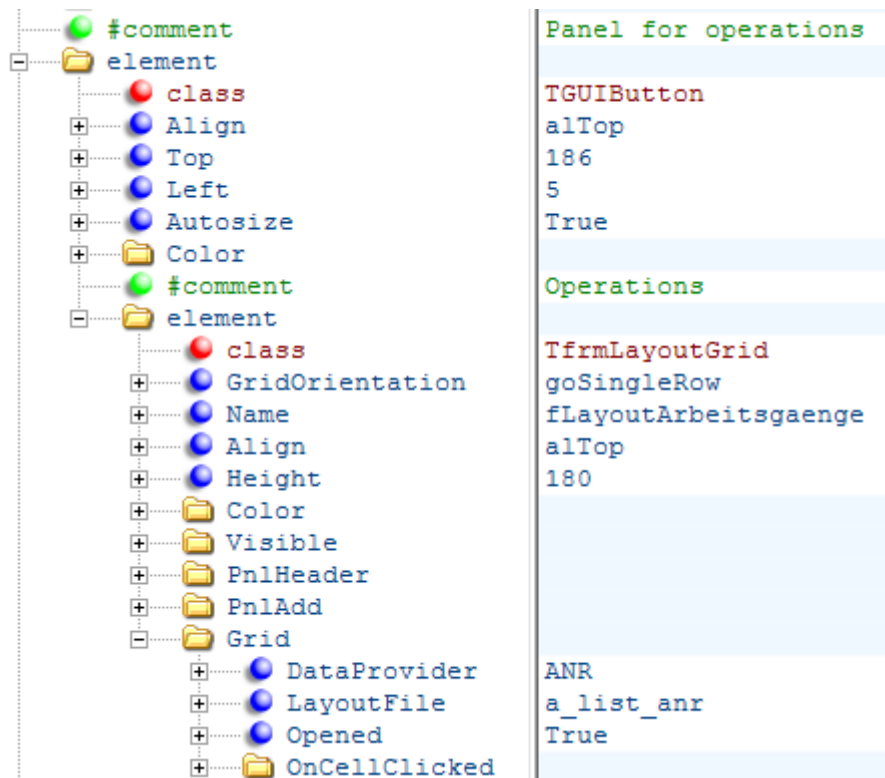


When you change the language during runtime using the flag in the status bar, the currently displayed layout is loaded. Changes of the layout in the main view are then taken over.

1.5 Configuration of lists

The configuration of lists is explained using the list of the logged on operations in the layout **I_main.xml**:





Lists of operations have their own panel in order to keep their position if operations of aggregates (a line is separated) are hidden.

The class *TfrmLayoutGrid* is responsible for the list display.

The settings below *PnlHeader* specify if and how a heading is displayed above a list.

The settings below *PnlAdd* define the button to create a new entry. In the above example, it's the button containing a "+"-symbol .

In the *Grid* area the data source (*DataProvider*) is set for the list. *LayoutFile* specifies which file defines the display of an element in the list. Below *OnCellClicked* you control what happens if you click an element in the list.

1.5.1 Filtering the displayed elements in the user interface (as of version 8.2.1.1)

The displayed lists can be filtered in the user interface. A text field must be included in the header of the list of type *TfrmLayoutGrid*.

The search syntax is equal to the search syntax of the lists in the dynamic dialogs.

A search field is integrate to a list of the type *TfrmLayoutGrid* eingebunden:

```

<element class="TfrmLayoutGrid">
  <SearchPanel>
    <Settings>
      <EnableSearch>true</EnableSearch>
      <SearchType></SearchType>
      <ExecuteEvent>return</ExecuteEvent>
      <SearchFields>ANR|AGNR</SearchFields>
      <SearchControlWidth>50</SearchControlWidth>
    </Settings>
  </SearchPanel>
  <SearchPanelPosition>header</SearchPanelPosition>
  ...
</element>

```

Details on the properties

Properties	Description
Settings.EnableSearch	Display search panel
Settings.SearchType	Currently, only text is possible. Describes the visual search component.
Settings.ExecuteEvent	Event that should trigger the search. The following configurations are possible: KeyDown The search is performed immediately on pressing the key. Compared to the following configurations, the above configuration costs a lot of time and should only be used for small data quantities. Button The search is performed by clicking an additionally displayed button. Return (Extension of "Button") The search can also be performed by pressing the return key.
Settings.SearchFields	Fields in the data source where you want to find the text you entered.

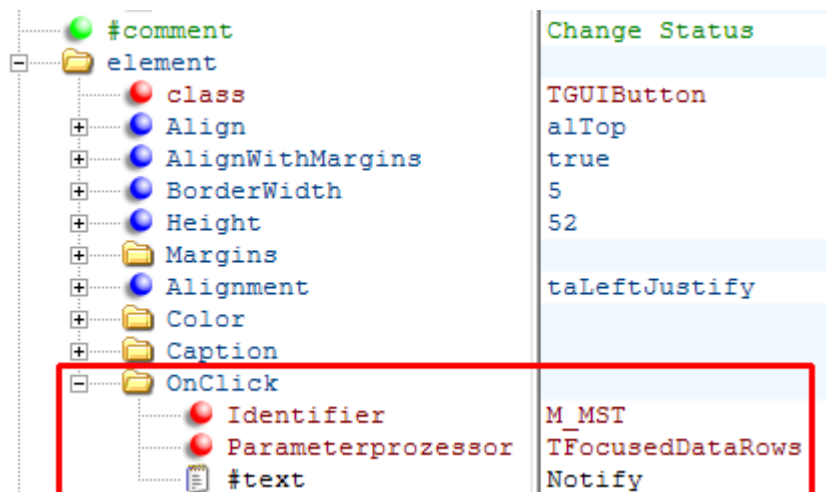
Settings.SearchControlWidth	Width of the input text box
TfrmLayoutGrid.SearchPanelPosition	Search panel position row: A row above the cells header: In the header of the TfrmLayoutGrids



If the search line is displayed, the list of type *TfrmLayoutGrid* requires more space in height. This space is at the expense of the tiles showing the data in the view. It is therefore recommended that you also change the height of the tiles with your data when using this functionality.

1.6 Request dynamic dialogs

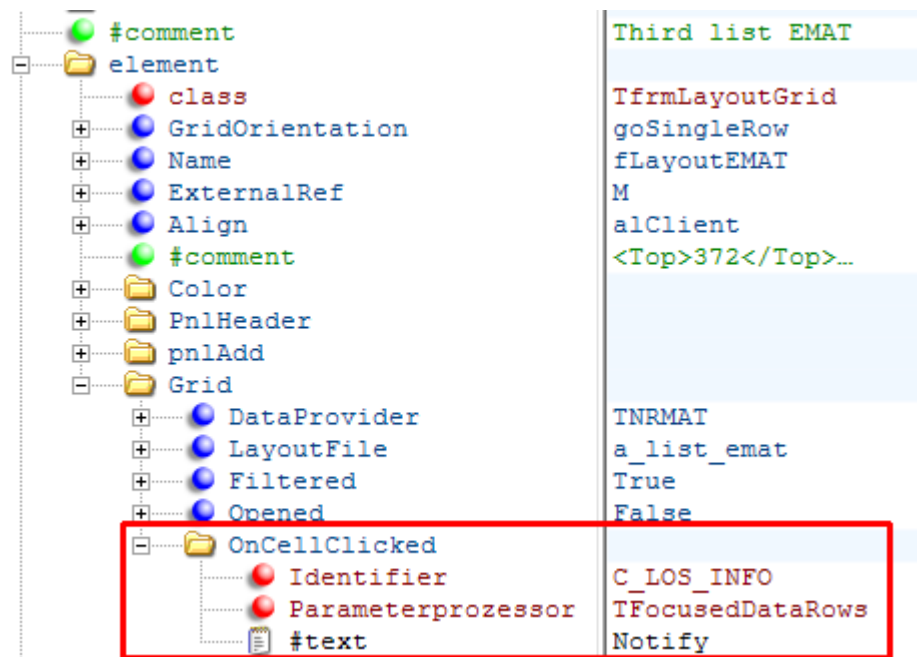
You can store an action in the individual fields and elements in order to perhaps request a dynamic dialog. In case of a button which is included on the left hand side in many layouts, you can enable this by using the entry *OnClick*:



This example shows the button "Change status" in the layout *I_mnr.xml*.

The entry *Identifier* defines the dynamic dialog to be requested. Both other entries are constant.

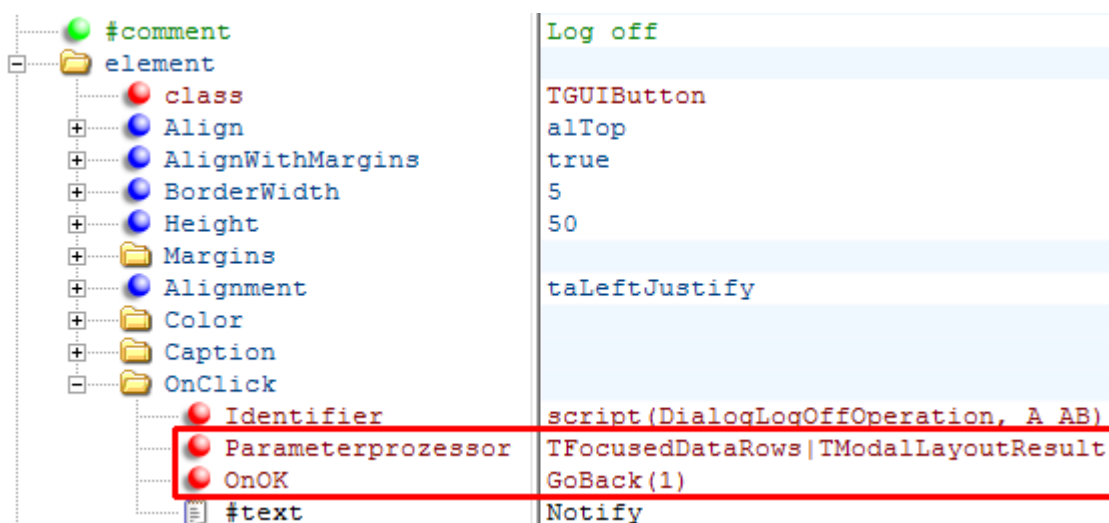
In a list of the class *TfrmLayoutGrid* containing several objects, the entry is called *OnCellClicked* and affects the elements below:



This example shows the request "MES Batch information" when selecting the input material in the layout *I_main.xml*.

1.6.1 Return after a dynamic dialog

After the execution of a dynamic dialog, you return to the layout where the dialog has been requested from. Alternatively, you can leave this layout and return to the previous layout which is normally the main view. Once an operation is logged off, it makes more sense to return to the main view than still displaying the data of a logged off operation.



A script request is stored in the setting *Identifier*. Depending on the workplace settings, the script request controls which dynamic dialog is requested. The first parameter specifies the script to be run and the second parameter the default value which is used if the script does not exist.



You then return to the previous view no matter if the dialog was completed or not, if errors occurred or if the dialog was interrupted without a change.

1.7 Positioning

The individual elements in the layout are arranged in a tree structure. The positioning of a subordinate item is always done in relation to the superior one (folder).

If a field shows a description and a corresponding value, the position of the description and the value refers to the top left corner of the field.

There are two ways to specify the position of the information. They are described in the following chapters.

1.7.1 Fixed positioning

Here, the position and the size of an element is specified with the following properties:

Top

Distance from the top

Left

Distance from the left

Height

Height of the element

Width

Width of the element

Properties can be found below the entry *control*.

The following example shows a logged on person for **a_data_pnr.xml**:

#comment	Person
element	
class	TGridItemControlPanel
control	
Top	0
Left	0
Width	190
Height	45
Color	
#comment	Label Person
element	
class	TGridItemLabel
DataFieldName	
control	
Top	5
Left	5
Caption	
#comment	PNR
element	
class	TGridItemLabel
DataFieldName	PNR
control	
Top	21
Left	5
Font	

The property *Alignment* also specifies if the element is positioned towards the left (*taLeftJustify*), or the right (*taRightJustify*) or towards the center (*taCenter*). If no other property is explicitly set, the standard setting is left-aligned. The following example shows a position towards the right of the label *Group* of workplace data (**a_data_mnr.xml**):

#comment	Label Group
element	
class	TGridItemLabel
DataFieldName	
control	
Top	5
Left	180
Alignment	taRightJustify
Caption	

1.7.2 Dynamic positioning:

If the positioning is done dynamically then the elements adapt their position and size to the one's above or next to it. The property *Align* can set the following:

Align=aTop / Align=aBottom

This element takes over the upper or lower limit and the width of the superior element. The property *Height* specifies the height.

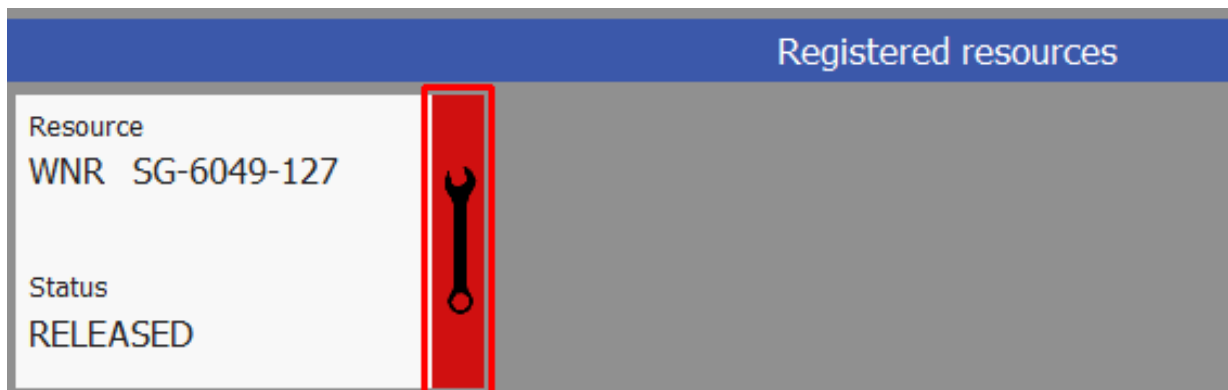
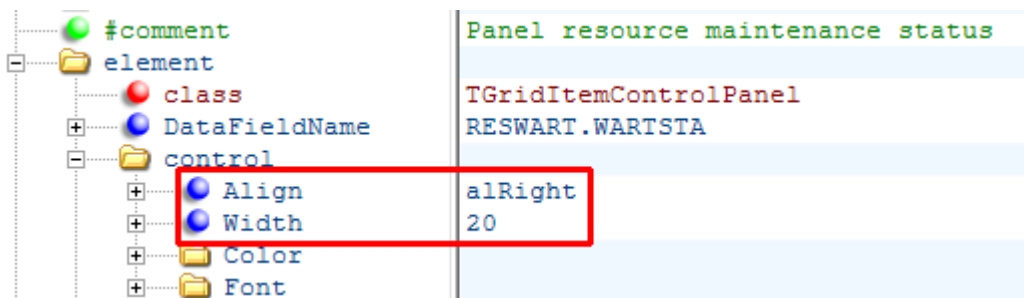
Align=alLeft / Align=alRight

This element takes over the right or left limit and the height of the superior element. The property *Width* specifies the width.

Align=alClient

Aligns itself to the complete space of the superior element.

The following example defines the area for the color display of the maintenance status in the list of resources (*a_list_res.xml*):

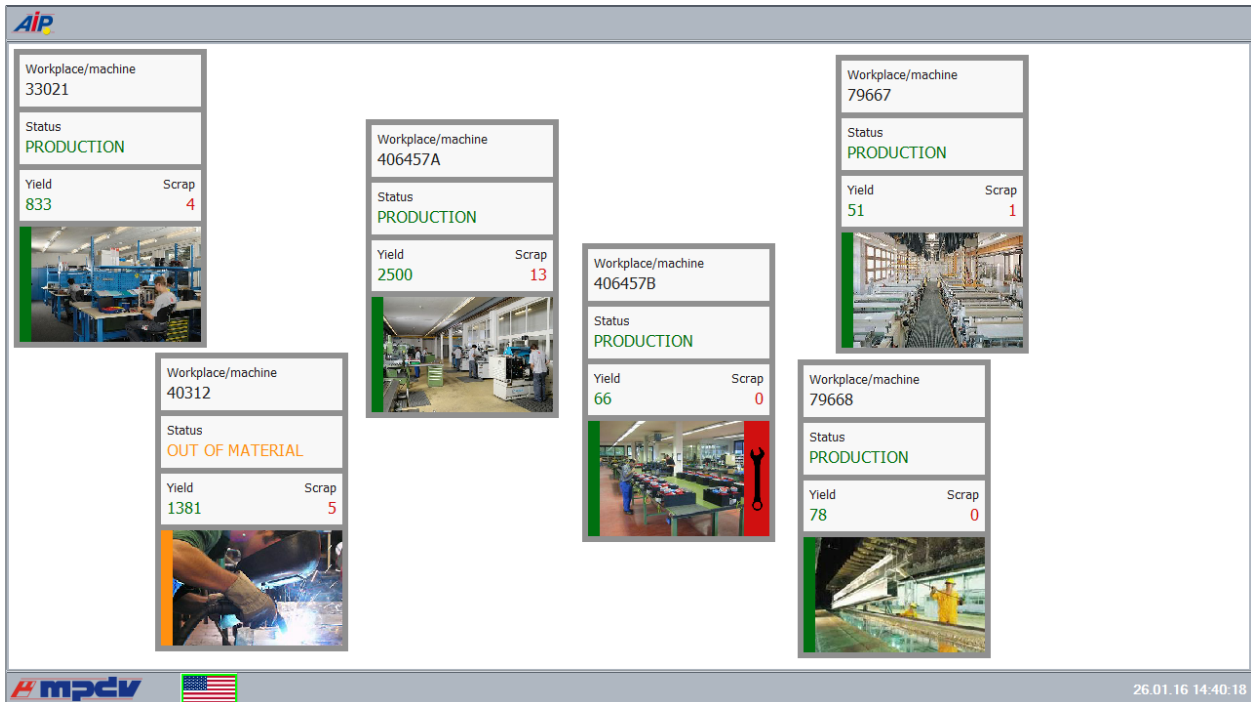


If neighboring elements have the same entry in the property *Align*, then they are displayed below or next to each other. This functionality is used in the button bars to request individual functions and ensures that there are no gaps if a function is hidden:



1.7.3 Positioning of workplaces in the icon view

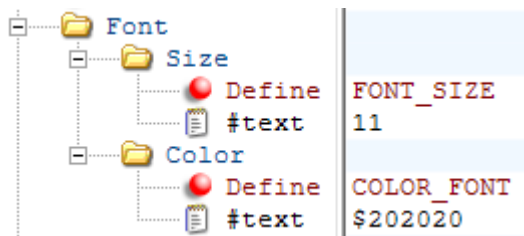
Positioning of individual workplaces in the icon view can be changed during runtime. You can start the design mode (password protected "mos6050") by double click the AIP icon in the top left corner. You can then position the workplaces by Drag&Drop. Double click the AIP icon to finish the design mode. The positioning of the workplaces is stored in the file *gui\p_view_mnr.xml*.



To reset the positioning, please delete the file `gui\p_view_mnr.xml`.

1.8 Text formatting

Text formatting in the GUI is performed using the entries below the field Font:



You can make the following settings:

Size

Set the font size

Color

Set the font color in reversed RGB notation



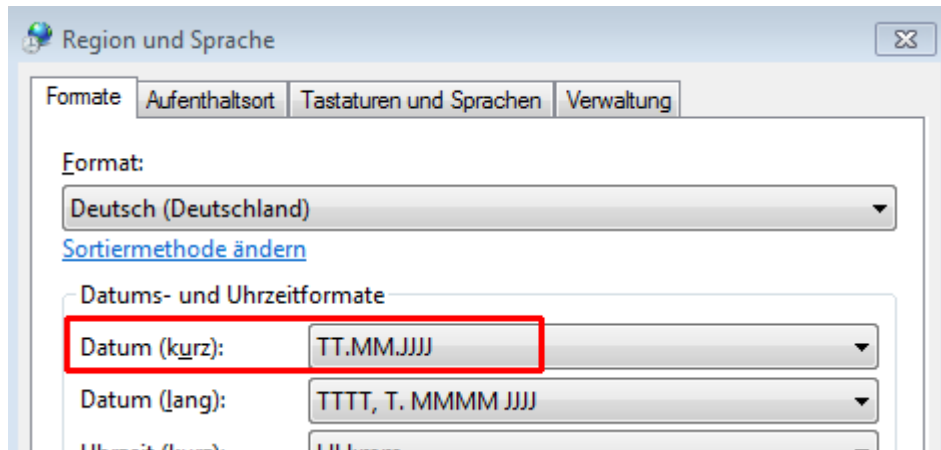
If the attribute *Define* is applied, the value entered in the field `#text` is not used. Instead the content of the entered constant is used (with both settings).

1.9 Formatting functions

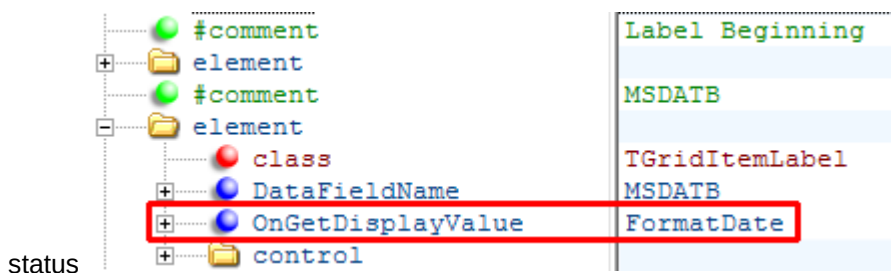
You can display data with the aid of different formatting functions:

FormatDate

This function sets a date depending on the date format *date (short)* set in the operating system:

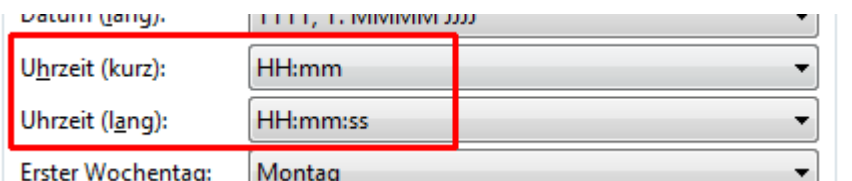


There is an example located in the workplace data (**a_data_mnr.xml**) for the start date of the current



FormatTime / FormatTimeLong

The function *FormatTime* sets the time depending on the time format *Time (short)* set in the operating system:



FormatTimeLong uses the format *Time (long)*.

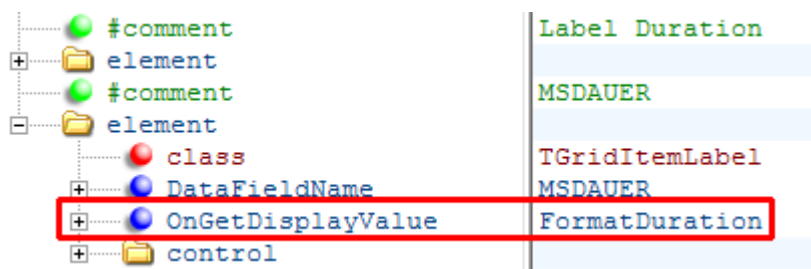
FormatDuration / ...

There are various functions to display *durations* in different output formats.

Function	Format
FormatDuration	Hours:Minutes
FormatDurationMMSS	Minutes:Seconds

FormatDurationHHMMSS	Hours:Minutes:Seconds
FormatDurationHHII	Hours,decimal hours (Industrial minutes)
FormatDurationHHIII	Hours, decimal hours (3 decimal places)
FormatDurationMMII	Minutes,decimal minutes

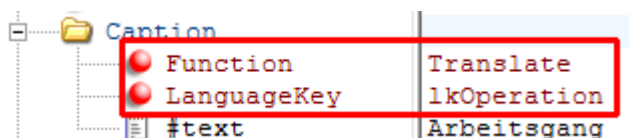
You can find an example for the output of a duration in workplace data(a_data_mnr.xml). It states the duration of the current status :



1.10 Multilingualism

AIP2 uses just like the AIP the Multilizer to translate text into another language. Language keys with the prefix "lk" are used for the new GUI. German texts without the prefix "lk" are also processed if they are included in the mld file.

Text for translation can be added using the function "Translate" and the entry "LanguageKey" (in accordance with the language set).



This example contains a German text "Arbeitsgang" (operation) which does not affect the processing. The text is replaced by the translated text using the language key during runtime.

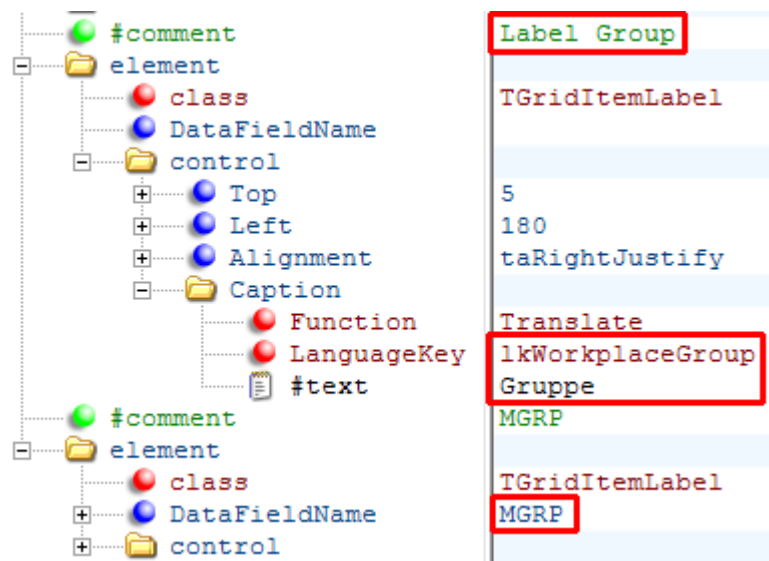
1.11 Examples / exercises

This chapter shows customization options of the layout using examples.

1.11.1 Change existing fields

Replace the field "group" with "cost center" in the displayed workplace data.

Workplace/machine				
Workplace/machine 33021	Short name WPL02	Group FINISH	Status PRODUCTION	
	Cycles 0	Start 18.08.2015	Duration [hrs:min] 06:00 3872:41	
	Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4



You need to change the entries for "label group" and "MGRP" (machine group) in the parameter **a_data_mnr.xml** as follows:



Entries with the description "#comment" are comments which do not affect processing.

The first element in the red frame shows the description above the data. The language key `IkWorkplaceGroup` is replaced by `IKCostCenter`. You can directly insert the text in the field `"#text"` if there is no language key available for the description. Both entries `"Function"` and `"LanguageKey"` must be deleted in this case.

The description is displayed towards the right hand side at position 180 (`"Left": 180; "Alignment": taRightJustify`).

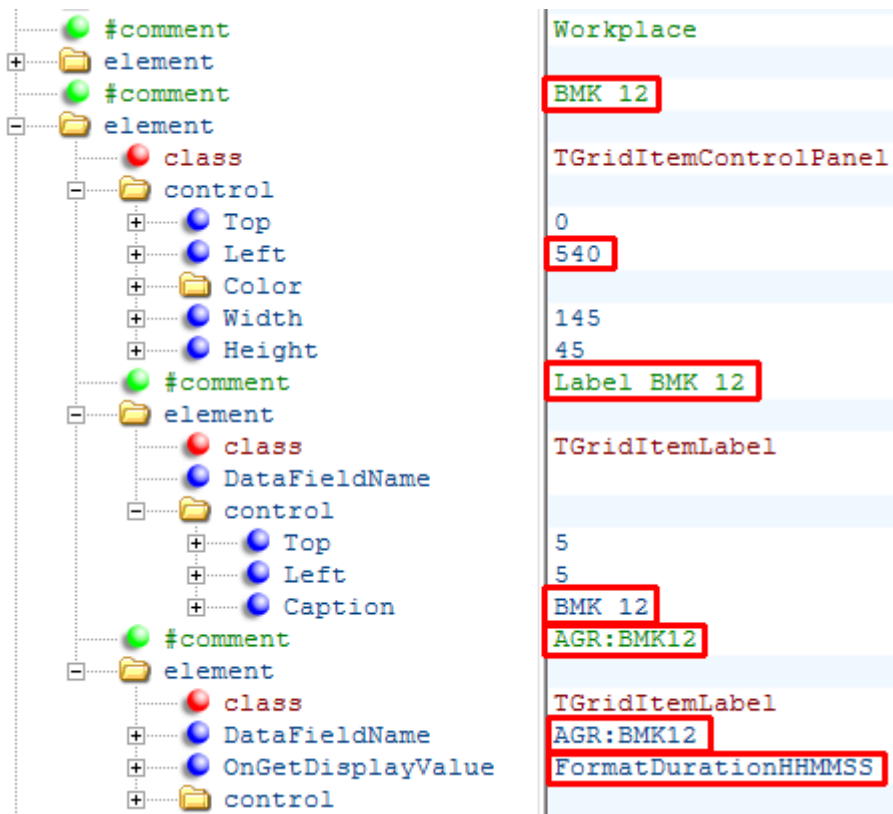
The second element is responsible for the display of the data field. Change the entry `"DataFieldName"` from `MGRP` to `KST`.

Workplace/machine				
Workplace/machine 33021	Short name WPL02	Cost center 4500	Status PRODUCTION	
	Cycles 0	Start 18.08.2015	Duration [hrs:min] 06:00 3872:43	
	Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4

1.11.2 Add a new field

Display the duration booked on RPA 12 to the right of the status display.

This is done by copying the element with the comment *Workplace*. This element specifies the light gray space and includes 2 other elements including name and data.



The comment located above is also copied and changed on BMK12.

The position of the light gray area ("left") is made up of position ("left") and the width of the element *Workplace Status* plus a distance of 5 dots ($345 + 190 + 5 = 540$). Both fields are located below the entry "Control".

The elements *Label BMK12* and *AGR:BMK12* are located on the light gray area. The position of both elements ("Top" and "Left") refer to the top left corner of the light gray space.

The entry "Caption" specifies the displayed text. Here, language keys have not been used so the text is not translated.

The entry *DataFieldName* below the comment *AGR:BMK12* was changed to the field name *AGR:BMK12*.

The formatting function *FormatDurationHHMMSS* shows the duration in hours:minutes:seconds.

Workplace/machine				
Workplace/machine 33021	Short name WPL02	Cost center 4500	Status PRODUCTION	RPA 12 5:00:00
	Cycles 0	Start 18.08.2015	Duration [hrs:min] 06:00 3872:47	
	Target cycle 20,000	Actual cycle 0,000	Yield 833	Scrap 4

1.11.3 Add user fields

Add a user field to the interface by adding a new field as described in the previous chapter. Use the the acronym of the user field (e.g. ANR_FU_65).



If you want to format user fields for dates, times, or durations, you can use the formatting functions. See section "1.9 Formatting functions".

Load user field with cataiplay.ini



Note that the AIP lists that serve as data providers do not contain user fields in the standard system. In order for the user fields to be added to the list, you must configure it in the **ctaiplay.ini** file. As long as the user fields in the **ctaiplay.ini** file are not configured correctly, they remain empty on the user interface.

In the customer-specific terminal directory (e.g. if user fields are to be added at terminal group level: \mip\<systemnr>\custom\aip2\tgrp_xxx\) a **ctaiplay.ini** is created, which contains the different section.

- Activate the additional loading of the user fields for operation- or order-related XML files in the section [Custom Userfields ANR].
- Activate the additional loading of user fields for machine-related XML files in the [Custom Userfields MNR] section.

You can find examples on how to do it further along.

The activated fields are then available in all XML files connected to the DataProvider ANR or MNR.

Available user field in machine and order lists

All identifiers of user fields and also other fields that can be reloaded are located in the headers.dat file in the "spool" directory of the terminal. It consists of four lines:

Start of the row	Content
10 ...	Machine list: Fields that are always included in the list.
*10 ...	Machine list: Fields that can be reloaded.
11 ...	Order list: Fields that are always included in the list.
*11 ...	Order list: Fields that can be reloaded.

The following user fields can be reloaded:

Machine list

FU:1 to FU:66

Machine user fields

Order list

ANR_FU_1 to ANR_FU_66

Operation user fields

AUNR_FU_1 to AUNR_FU_66

Order user fields

MNR_FU_1 up to MNR_FU_66

Machine user fields

VERARBCODE_FU_1 up to VERARBCODE_FU_66

Processing code user fields

AGR_FU_1 to AGR_FU_66

User fields of the operation status (cannot be used in the standard system, reserved for Customizing!)

Example 1: User fields in the operation list

- User field 1 of the operation should be entered in the order list with the name " Order date ".
- User field 66 of the machine with the name "My long user field" should be added to the order list.

Field definition in section [Custom user fields ANR]

```
[ Custom userfields ANR ]
```

```
GRID_LIST_TYP=ANR
```

```
; additional fields of the order list
```

```
ANR_FU_1= ; User field 1 of operation, MyDate FU:1 [operations list]
```

```
MNR_FU_66= ; User field 66 of machine, My long user field [operations list]
```

Example 2: User field in the machine list

User field 66 of the machine with the name "My long user field" should be added to the machine list.

Field definition in the section [Custom user fields MNR]

```
[ Custom userfields MNR ]
```

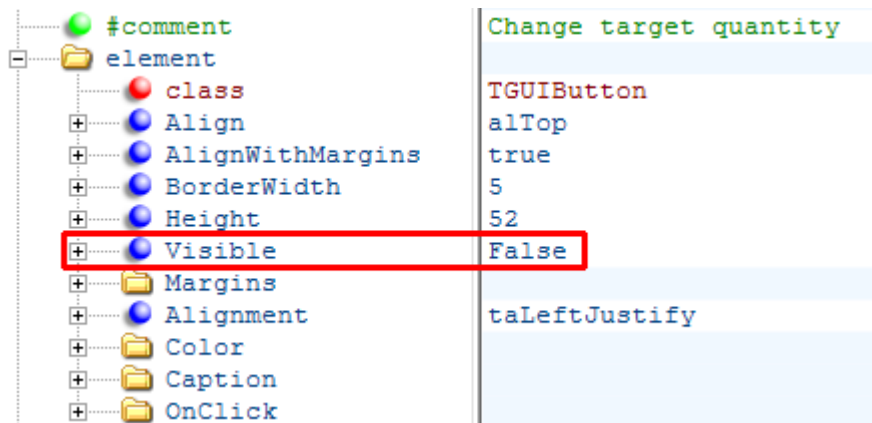
```
GRID_LIST_TYP=MNR
```

```
; Additional fields in the machine list
```

```
FU:66= ; User field 66 of machine, My long user field [machine list]
```

1.11.4 Remove button

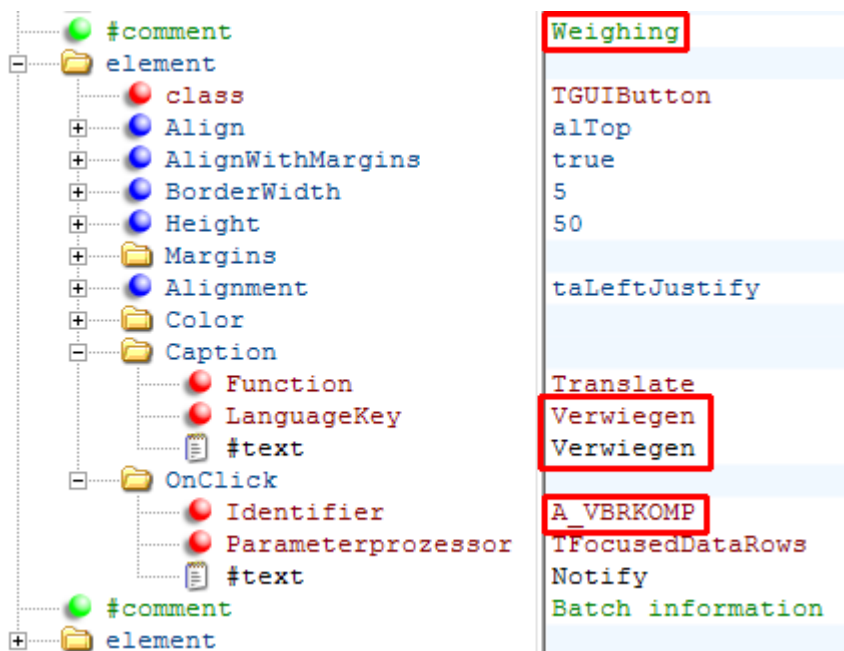
Delete the button *Change target quantity* from the operation layout (**I_anr.xml**).



The new entry `Visible=False` hides the button. Optionally, you can also delete the comment and the element.

1.11.5 Add button

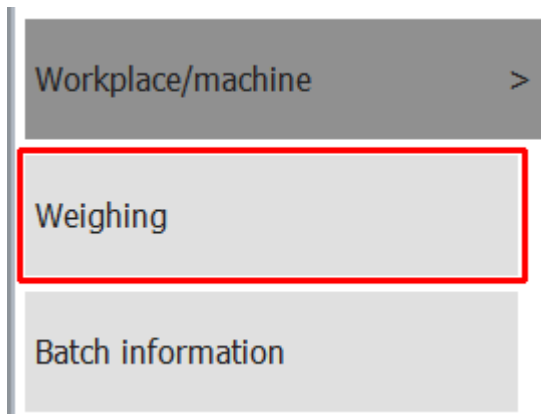
You want to add a new button "Weighing" in the layout for "input material", "output batch" and (**I_mat.xml**).



First of all, copy an existing button including the corresponding comment. In this case the button "Batch information" was copied. Change the comment in order to easily find the new button in the list of elements.

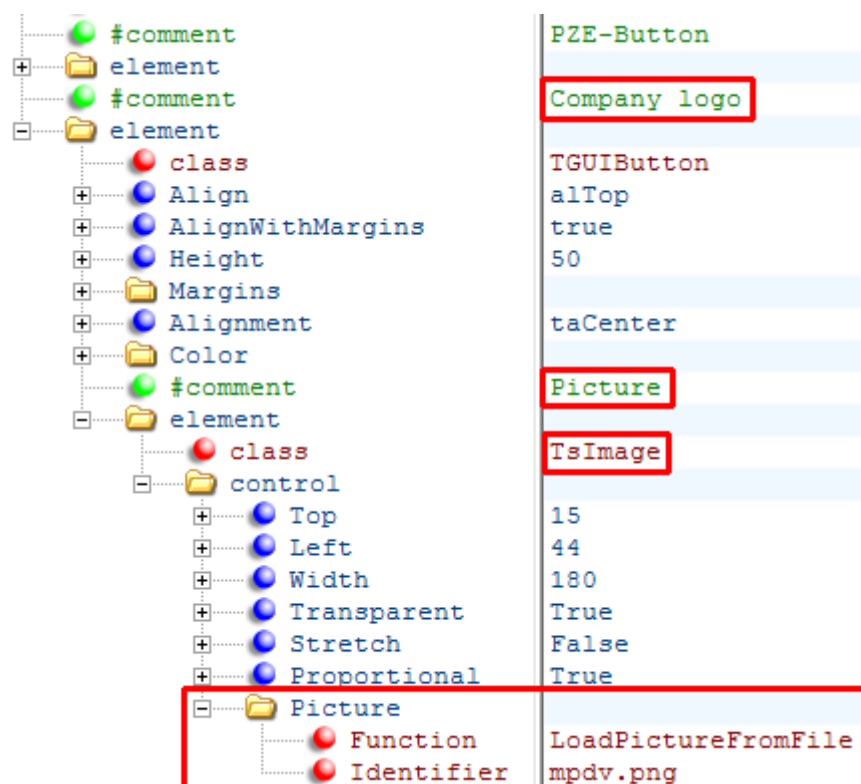
The entry "Caption" specifies the displayed text. In the example, the English text "Weigh" is used as language key.

The "Identifier" specifies which dynamic dialog is requested.



1.11.6 Integration of a picture

The task is to have a logo displayed in the main view (`I_main.xml`) below the button "PZE".



Copy the button "PZE" and the comment. Change the comment.

As the button has no labeling, delete the entry "Caption". Also delete the entries "Visible" and "OnClick".

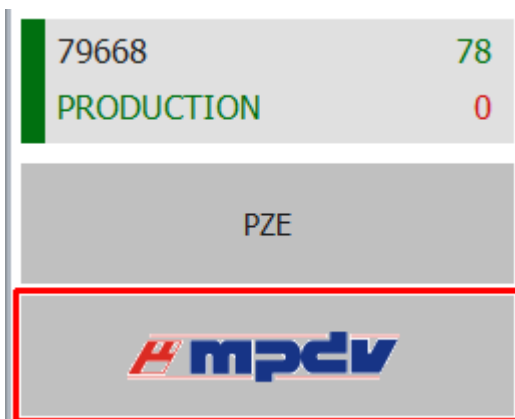
You need a new element of the class "TsImage" in order to display the new picture. This element was copied from the staff list (`a_list_pnr.xml`) and has the class "TGridItemImage", as it is not located on a button but in a list. Change the class to "TsImage" after copying. Delete the entry "Visible".

The file name of the picture is entered in the field "Identifier" of the entry "Picture". There are two options to load the picture:

- **LoadPictureFromFile** reads the file from the spool directory.
- **LoadPictureFromAIP** uses pictures included in the AIP2 in the file "pict.zip" or "pict_cust.zip". This information is more efficient as these picture are stored in a buffer. This method only supports images of type PNG and BMP.

Different settings are available to display the picture.

- - **Transparent** – For example, PNG files support transparent areas where the background of the picture is visible. Functionality can be switched off using the value *False*.
- - **Stretch** specifies if the picture is shown in its original size (value *False*) or if *Height* and *Width* are adjusted (value *True*).
- - **Proportional** controls whether the ratio of the width of the image and the height of the image is maintained (value *True*) or not (value *False*) when the image size is adjusted to the specified *Height* and *Width*.



1.11.7 Change quantity format

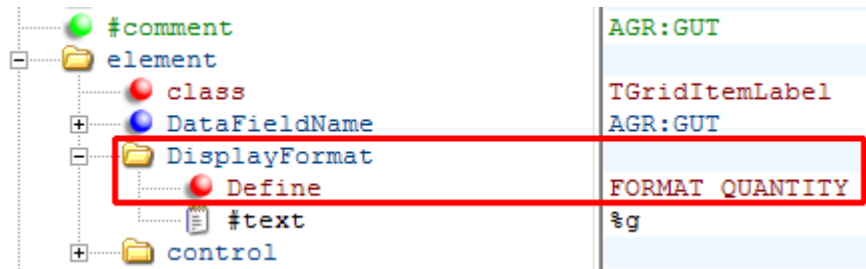
Generally show the quantity format with 2 decimal places.

The quantity format is configured in the file **globaldefines.xml** using the constant *FORMAT_QUANTITY*:

```
#comment
FORMAT_QUANTITY      Format for quantities (e.g. %g, %0.2f)
                      %0.2f
```

The value "%0.2f" ensures that quantities are displayed with 2 decimal places.

You can find an example for this constant (Define) when yield is displayed for data of the workplace (**a_data_mnr.xml**):



Workplace/machine					
Workplace/machine 33021	Short name WPL02	Group FINISH	Status PRODUCTION		
	Cycles 0	Start 18.08.2015 06:00		Duration [hrs:min] 3872:56	
	Target cycle 20,000	Actual cycle 0,000	Yield 833,00	Scrap 4,00	

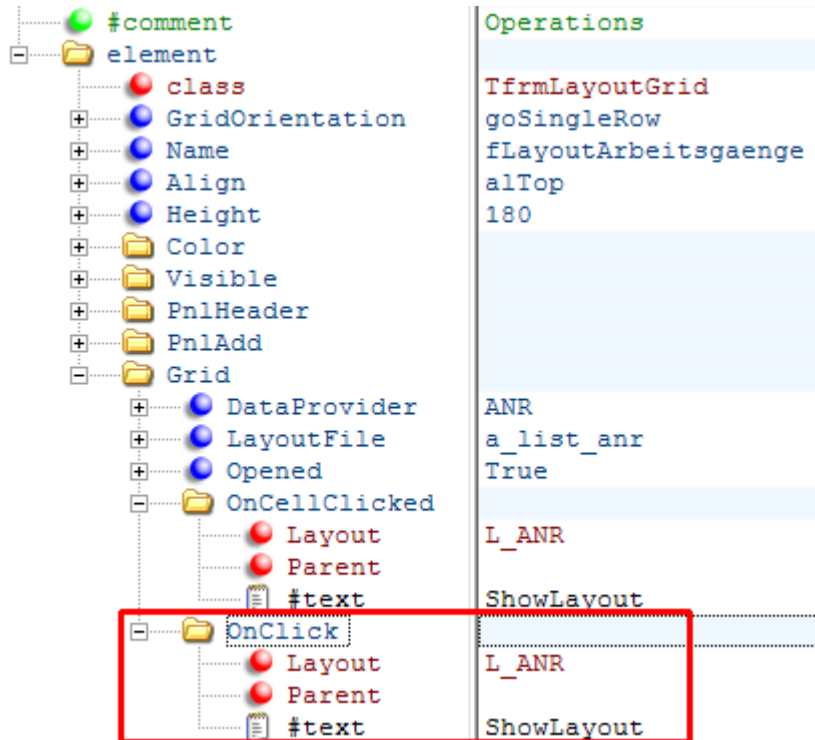


The set format only affects configured layouts and not dynamic dialogs.

1.11.8 Postings for operations not logged on

The workplace configuration in the MOC has a button called "Posting of operations not logged on". If this button is activated, you can interrupt or log off the operations not logged on (posting to the server). You have to extend the configuration if you would like to carry out these postings in the AIP2.

The operation layout only opens in the AIP by using the buttons *Interrupt* and *Logg off* if you select the logged operation. If the following extension in the file `I_main.xml` is carried out, this layout also opens if you click the empty space in the list of operations.



The dynamic dialogs must also be customized in order to interrupt and logg off operations. Unless so-called simple dialogs are used.

1.12 Index

#comment.....	31
#text.....	10, 28, 32
ActionOnLostFocus.....	19
alBottom.....	26
alClient.....	26
Align.....	26
Alignment.....	26
alLeft.....	26
alRight.....	26
alTop.....	26
Caption.....	33, 35
Color.....	28
control.....	25
DataFieldName.....	32, 33
DataProvider.....	21
Define.....	28, 37
Defines.....	5
Extention.....	10
Font.....	28
FormatDate.....	29
FormatDuration.....	29
FormatDurationHHII.....	30
FormatDurationHHIII.....	30
FormatDurationHHMMSS.....	30, 34
FormatDurationMMII.....	30
FormatDurationMMSS.....	30
FormatTime.....	29
FormatTimeLong.....	29
Function.....	32
Grid.....	21
Height.....	25, 26
Identifier.....	23, 24, 35, 36
laFree.....	19
laHide.....	19
LanguageKey.....	30, 32
LayoutFile.....	21
Left.....	25
LoadPictureFromAIP.....	37
LoadPictureFromFile.....	36
OnCellClicked.....	21, 23
OnClick.....	23
PnlAdd.....	21
PnlHeader.....	20
Proportional.....	37
ScriptName.....	10
Size.....	28
Stretch.....	37
taCenter.....	26
taLeftJustify.....	26
taRightJustify.....	26, 32
TfrmLayoutGrid.....	20, 23
TGridItemImage.....	36
Top.....	25
Translate.....	30
Transparent.....	37
TsImage.....	36

Visible	34
Width.....	25, 26